

Introduction

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COURSE ORGANIZATION AND RESOURCES

This course has multiple learning objectives. As you approach your senior year and prepare for independent practice as veterinarians, our goal is to equip you with the skills to efficiently find information on these species in a clinical setting while also providing you with foundational knowledge of small mammal medicine. In addition, our case-based examination should reflect the manner that some NAVLE questions are formulated, but most importantly reflects knowledge application to clinical scenarios.

Learning and studying to succeed in this class and prepare for veterinary practice may feel like separate goals. But we have worked very hard in this course to make it as clinically relevant as possible in combination with the **Companion Exotic Animal Medicine and Surgery (CEAMS) clinical rotations**. We hope this information will be sufficient to acquire an entry-level knowledge necessary to care for these species upon graduation.

This course has about 11 instructors. Therefore, teaching styles and materials may vary between instructors, but overall the course is fairly comprehensive touching on most topics relevant to small mammal practice. These topics are divided into lectures, discussions, and practical laboratories while limiting redundancies of information. The discussion and labs are aimed to make you apply your knowledge. Some concepts are only taught in labs or discussions such as physical examina-

tion and sample collections.

On top of the lectures, discussions, and labs, there are a few “homeworks”. There are **“additional readings” or other module activities** depending on the instructor that are optional, but cover important aspects of the lectures that could not be covered elsewhere because of time constraints. There are also lots of cool videos to look at. In addition, please review the relevant information prior to the labs so you come prepared and optimize your time with the animals.

This course includes **21 lectures** and, to limit redundancies, some information is also found in other courses (such as dermatology in the skin block for instance). So if you think something is missing, it is probably because it is already covered some-

where else as part of a comparative medicine approach. Anything you feel would improve the lectures, feel free to let Dr. Beaufriere know (hbeaufriere@ucdavis.edu). You can also use the **discussion in Canvas** or reach out to the other instructors. Depending on the lectures, there are also some fun **self-learning activities** available.

There are **4 discussions**: diagnostic imaging, orthopedic, case studies, and a general discussion. These are case-based, so please review any materials posted on canvas in the discussion folder before.

There are **4 labs**: clinically relevant anatomy, physical examination and clinical techniques, clinical pathology, and dentistry. There are **many videos** to help prepare for the techniques taught in these labs, but also additional videos for techniques not covered in the labs, but still important for practice.

The best part of this course is obviously the **quizzes**



and exams, we know how much you love them! **Quizzes are 50% of the grade. There are 5 quizzes worth 10 points each: anatomy, gastrointestinal, urogenital, emergency/critical care, and neuromuscular.** Some are case-based and designed to help you prepare for actual clinical cases. You have **36h to complete each quizz and they are open book.** They may overlap a bit because it is only a 2-week course.

The final examination is the other 50% of the grade. It is composed of multiple choice questions, true/false questions, and short answers depending on the instructors. Questions could come from any of the materials in the class including lectures, discussions, labs and all associated required materials. **The exam will be open book** (you're welcome!!). It will be administered on **Examsoft using the non-secure mode** (so you can consult your course materials) but **restricting the use of internet (so download anything you may need before hand).** Reference intervals for blood work will be given with examination questions that need them. **There is a 1h discussion session for you to ask us questions before the examination.** Feel free to ask questions regarding this course or regarding small mammal medicine in general.

WHY SMALL MAMMALS?

Some of you may have a strong interest in small mammal medicine, while others may be uncertain or question the relevance of learning this material. There are many reasons why this course is now included in the Small Animal Emphasis of the third-year DVM curriculum at UC Davis.

Small mammals are **very commonly kept as pets, especially rabbits, rodents, ferrets** and a few other exotic mammals. **The rabbit (or small mammals as a group depending on surveys) is now the third most popular mammal pet** with likely over 3 million rabbits being kept as pets in the US. With the population getting more urbanized, small mammals are poised to keep becoming more popular as they are easier to keep in apartments. This means that we need to have enough veterinarians to provide medical care for these pets. Many clinics see a bit of exotics. In addition, **the veterinary market for dogs and cats is saturated in some areas and adding exotic animals to a clientele allows to have a competitive edge** and also cater for clients with multiple animal species. It is also important for overall captive animal welfare that veterinary care

availability matches pet ownership demographics.

Exotic animal medicine (a branch of zoological medicine) also includes birds, reptiles, invertebrates, and fish, but **it is easier to learn how to treat small mammals than it is to learn the medicine of non-mammalian species** as you already know a lot about mammalian anatomy and lots of other things that can easily be extrapolated (especially for ferrets as it is a carnivore like the dog and the cat). On the other hand, the learning curve is steeper to become competent in avian and reptilian medicine (see **VET437**). Thus, adding small mammals to a companion animal practice is much easier than adding other species of exotic animals.

It also brings variety to your practice and to your professional life as a whole so you don't end up just treating 1-2 species (remember the main difference between MDs and DVMs is that we treat more than one species).

In addition, in terms of mammalian biodiversity, **rodents are the most diverse mammal taxon** with more than 2000 species. Since there are about 6400 species of mammals, about a third of all mammals are rodents. This has implications if you are interested in the fields of zoological and wildlife medicine because you will definitely see rodents there (cool zoo rodents include capybaras, naked mole rats, all kinds of squirrels, prairie dogs, marmots, beavers, porcupines and so on). If you know the dentistry of rabbits and pet rodents, you know the dentistry of zoo rodents to some extent.

There is also **plenty of information on small mammal medicine and physiology from the laboratory animal field and we take advantage of that knowledge for our patients.** Rats, mice, hamsters, rabbits, and guinea pigs are used for all kinds of research. Ferrets are heavily used as laboratory animals as models for respiratory viruses (influenza, SARS-Cov2) and gastric ulcer research. Chinchillas have been used for ear and otitis research because of their large middle ear.

With that said, **laboratory animal medicine and exotic animal medicine have many differences so not all knowledge is transferable to our patients.** The animal breeds used in the laboratory are not the same as the ones kept as pets, so even some basic information such as blood work values cannot necessarily be directly extrapolated. Pets also tend to live longer, have different diets, be exposed to different infectious diseases than their laboratory counterparts, and have many diseases not seen in laboratory animals. We thrive to use pet-specific information first whenever we can.

Some of these species are also kept in non-pet situations. While these are not necessarily relevant to exot-

ic pet medicine (or sometimes in direct opposition), it is still worth being aware of these industries. Rabbits are bred for meats for instance. The Californian rabbit breed is one of the most popular meat breeds of rabbits. Angora rabbits are raised for wool production. Ferrets have also been used to hunt rabbits for centuries (called ferreting, see picture below) in Europe. Minks and chinchillas are farmed for their fur.



We hope you'll enjoy this course and get the most out of it and that it will stimulate you to provide care for these wonderful pets.

SPECIES AND TAXONOMY

While we won't ask any questions on that on the examination, it may still be important to know some basic taxonomic facts about these species. Rabbits are in their own order, the **Lagomorpha**, which contains also pika (family Ochotonidae) and rabbits and hares (family Leporidae). **The pet rabbit (more than 300 breeds) is coming from the European rabbit.** The wild rabbit is a digging species and lives in large networks of burrows. While it is an invasive species in many world regions, it is declining in its original range due to viral diseases such as myxomatosis and viral hemorrhagic disease. It is a different species from North American rabbit species, which don't dig burrows (such as the cotton tail rabbits). Like some dog breeds, **some rabbits breeds have been selected for traits that interfere with their health** such as lop-eared rabbits, which are extremely prone to otitis or dwarf rabbits, which are prone to dental disease.

There are many species of rodents commonly kept as pets. Hamsters, gerbils, rats, mice originally come from Europe/Asia. Rats and mice have co-existed with humans for a while. Chinchillas and guinea pigs are South American rodents. **Pet species of rodents also have various breeds.** For instance, guinea pigs have about a dozen common breeds which mainly differ by their hair coats such as the American guinea pig, the most com-

monly seen breed, or the hairless skinny pig. Chinchilla breeds mainly differ by their color pattern such as the standard grey, the black velvet (chinchilla on page 1), the white, beige, brown or purple.

NAME	SCIENTIFIC NAME	FAMILY	ORDER
Rabbit	<i>Oryctolagus cuniculus</i>	Leporidae	Lagomorpha
Guinea pig	<i>Cavia porcellus</i>	Caviidae	Rodentia
Chinchilla	<i>Chinchilla lanigera</i>	Chinchillidae	Rodentia
Degu	<i>Octodon degus</i>	Octodontidae	Rodentia
Syrian hamster	<i>Mesocricetus auratus</i>	Cricetidae	Rodentia
Dwarf hamsters	<i>Phodopus spp.</i>	Cricetidae	Rodentia
Rat	<i>Rattus norvegicus</i>	Muridae	Rodentia
Mouse	<i>Mus musculus</i>	Muridae	Rodentia
Mongolian gerbil	<i>Meriones unguiculatus</i>	Muridae	Rodentia
Ferret	<i>Mustela furo</i>	Mustelidae	Carnivora
African hedgehog	<i>Atelerix albiventris</i>	Erinaceidae	Eulipotyphla
Sugar glider	<i>Petaurus breviceps</i>	Petauridae	Diprotodontia (marsupial)

Rabbits are not closely related to rodents phylogenetically so they have many anatomical and physiological differences. **The lagomorphs and rodents together form the Glires clade.** For practical purpose (not taxonomic based, but based on rostral morphotypes), **rodents can be further divided into hystricomorph (guinea pig, chinchilla, degu), myomorph (mouse, rats, hamsters), and sciurimorph (squirrels, marmots) rodents.** This has clinical implication as hystricomorph rodents are treated more like rabbits as they have similar dental and gastrointestinal disorders.

While most of the small mammal pets are glires, there are a few exception. **The ferret is a carnivore that is a descendant of the European polecat (*Mustela putorius*),** but is now considered a separate species. It belongs to the mustelids along with badgers, martens, otters, weasels, and wolverines. Two oddballs that we sometimes see are the **African pygmy hedgehog**, an insectivore from central and eastern Africa, and the **sugar glider**, a marsupial from Australia. Despite their popularity, less is known on how to best keep them in captivity, so many husbandry and dietary issues are seen.

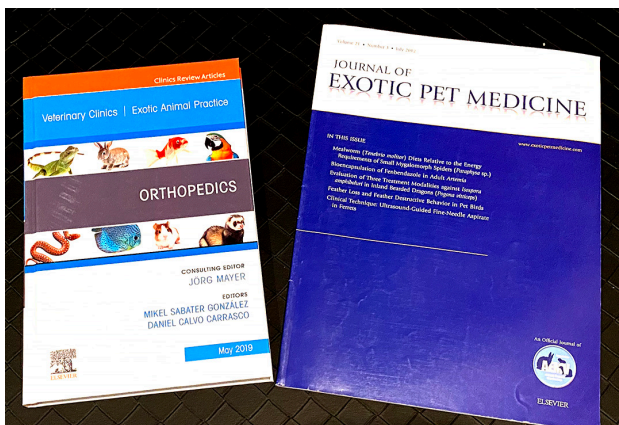
Unfortunately (or fortunately depending on how you look at it), **many small mammal species are banned as pets in California** including the ferret, sugar glider, hedgehog, degu, and gerbils.

RESOURCES

This section is aimed at summarizing the main sources of information related to small mammal medicine. You obviously already have access to great resources within this course, but this may help if you want to dig deeper on some concepts, read more on specific diseases, or just to know where to find information from different veterinary and non-veterinary sources.

Veterinary Literature

The amount of information on medicine and surgery of exotic species is growing by leaps and bounds every year. New textbooks are being published consistently as well as many scientific articles either **in general veterinary journals or in specialty journals**. Some interesting articles can also be found in **laboratory animal medicine journals**.



Small Mammal - Scientific Literature

General Journals

[American Journal of Veterinary Research](#)
[Journal of the American Veterinary Medicine Association](#)
[Journal of Small Animal Practice](#)
[Veterinary Record](#)

Specialty Journals

[Journal of Exotic Pet Medicine](#)
[Veterinary Clinics of North America: Exotics Animal Practice](#)
[Journal of Zoo and Wildlife Medicine](#)

Laboratory Animal Journals

[Journal of the American Association for Laboratory Animal Science](#)
[Comparative Medicine](#)

Specialty Small Mammal Books

Textbook of Rabbit Medicine
Ferrets, Rabbits and Rodents, Clinical Medicine and Surgery
The Laboratory Rabbit, Guinea Pig, Hamster and other Rodents
Current Therapy in Exotic Pet Practice
Dentistry in Rabbits and Rodents
Biology and Diseases of the Ferret
BSAVA Manual of Rabbit Medicine
5-Minute Veterinary Consult - Small Mammals

General Exotic Books with Small Mammal Sections

Exotic Animal Emergency and Critical Care Medicine
Exotic Animal Formulary
Exotic Animal Laboratory Diagnosis
Surgery of Exotic Animals
Veterinary Clinical Advisor: Birds and Exotic Pets

Many of these journals and books are available through our library including electronic versions. That's literally thousands of dollars at your fingertips. Download any chance you can!!



Associations and Online Resources

There are lots of pet owners' clubs, breeders, laboratories, and not-for-profit associations and organizations that are associated with small exotic mammals. Sometimes, useful information can be found on their website such as recommended vaccination protocols, breed standards, recommendations on husbandry and diet, and relevant news pertaining to these species.

Small Mammal Associations

Non-Veterinary

[House Rabbit Society](#)

[American Ferret Association](#)

[American Rabbit Breeder's Association](#)

[American Fancy Rat & Mouse Association](#)

[American Cavy Breeder's Association](#)

Veterinary

[Association of Exotic Mammal Veterinarians](#)

[American Association of Zoo Veterinarians](#)

[American Association for Laboratory Animal Science](#)

[Association of Sugar Glider Veterinarians](#)

Many review papers, conference proceedings on exotic animals, and topic articles can also be found online on **continuing education websites**. Some will even host webinars for continuing education. As some of these websites are associated with pet food companies, you should keep a critical mind when reading some of the articles for which there might be a conflict of interest.

In addition, there are a number of conferences for continuing education or to learn the latest trends and updates on small mammal medicine. Some of these conferences are specific to the field [one international conference in Europe (ICARE) and one in the USA (ExoticsCon)] while some are more generalistic. Nowadays, **most general veterinary conferences offer sessions on small mammal topics** as there is a huge demand for this from practicing veterinarians who also see dogs and cats.



ASSOCIATION
OF EXOTIC
MAMMAL
VETERINARIANS

EXOTICSCON

Continuing Education

Online Resources

[Veterinary Information Network](#) (VIN)

[LafeberVet](#)

[Oxbow Vet Connect](#)

Exotic Medicine Conferences

[ICARE](#) (international conference on avian, reptiles and exotic mammals, in Europe)

[ExoticsCon](#) (US)

General Conferences with Exotic Sessions

[VMX](#) (Veterinary Meeting & Expo)

[Western Veterinary Conference](#)

[Wild West Veterinary Conference](#)

[Pacific Veterinary Conference](#)

[ABVP symposium](#)

SPECIALIZATION

In North America, to specialize in small mammal medicine, you can either pursue species-based specialization through the [American Board of Veterinary Practitioners](#) - Exotic Companion Mammal (ABVP-ECM) or a discipline-based specialization through the [American College of Zoological Medicine](#) (ACZM) - Companion Zoological Animals (ZCA). ACZM residencies typically encompass more than small mammals while ABVP-ECM residencies are only focused on small mammals. ACZM-ZCA residencies tend to be in academic institutions while ABVP residencies tend to be in private referral practices.

BASIC EQUIPMENT

To treat small mammal patients, you need the appropriate medical equipment and supplies for sample collection, hospitalization, basic procedures, dentistry, and treatment. As these animals are small, miniaturized equipment is needed.

For physical examination, on top of general veterinary tools, you will need a **gram scale** (with a weight basket for the smallest animals) to weigh the animals, an oral speculum for oral examination of rabbits and rodents, a pediatric stethoscope for cardiopulmonary and GI auscultation, and towels for restraint.

For sample collection, smaller needles (27g to 22g) are needed as well as smaller syringes including tuberculin and insulin syringes. Mini culture swabs are also used

to obtain culture samples from narrow cavities such as the nasal cavity.

For hospitalization, **incubators** should be available that can also serve as oxygen cages and nebulization chambers. A range of dietary items should be kept in the hospital such as several types of grass hay for rabbits and rodents, pelletized diets for the range of small mammals seen, and a supply of fresh greens if possible. Critical care diets should also be maintained such as powdered hay diet from brands such as Oxbow and Emerald and carnivore liquid diets.



For diagnostics, the usual veterinary equipment can be used such as digital radiography and ultrasounds with higher frequency probes than for dogs and cats (especially for rodents). A small fine clipper should be acquired for shaving, especially for rabbits as their skin is thin and prone to tearing.

Dentistry is a big part of what you do in rabbits and rodents and the appropriate instruments should be available. State-of-the-art dentistry in rabbits and rodents now involves the use of endoscopic equipment to magnify structures in their small mouth, record lesions, and assist with treatment. See [Dentistry lecture](#) for more information. Equipment for dentistry in ferrets is the same than for dogs and cats.

In terms of anesthesia, most of the equipment used in dogs and cats can be used on rabbits, ferrets, and rodents. Smaller endotracheal tubes are used, typically uncuffed as their diameter is smaller and they are easier to place. Large IV catheters can also be used in smaller rodents. **Supraglottic airway devices** have been made for rabbits (V-Gel, single use) and can be used in place of intubation for simple procedures or for CPR when quick intubation can be hard to perform. The V-gel size R1 can also be used in African hedgehogs. Anesthetic monitors need to be adapted to small patients with small tidal volumes so capnographs need to be microstream. Digicare also has multiparameter monitors for exotic animals. Ventilators that we use are typically pressure-controlled ventilators so we can ventilate tiny animals. The Vetronic SAV04 is a commonly used one. See also the [Anesthesia lecture](#) for more information.

For surgery, similar equipment is used as in small animal surgery. The main difference is that microsurgical instruments are more commonly used (see [Soft Tissue Surgery lecture](#)).

Rabbit/Rodent Dentistry Equipment

Endoscopic equipment

- Short rigid endoscope: videotoscope or other short-length endoscope, diameters of 2.7 to 5 mm
- Endoscopic camera
- Light cable
- Endoscopic recording system and tower (portable towers are cheaper such as the Storz Telepack)

Positioning

- Rabbit/rodent dental table with mouth gag
- Cheek spreaders (different sizes)

Trimming tools

- Low speed dental unit
- Straight hand piece dental drill
- Diamond-coated dental burs
- Soft-tissue protector (goes around the burr)
- Cutting disc or angled high speed drill (for incisors)

Dental instruments

- Rabbit/rodent incisor luxator
- Rabbit/rodent molar luxator
- Small all-purpose dental scaler
- Cheek teeth extraction forceps

Other

- Cotton tip applicators
- Tongue depressors
- Local anesthetic gel
- Anesthetic equipment

Don't forget to take the CAPE rotation !!

